

## A Written de Sitter–Unruh Modified-Inertia Action (v6)

**The passive-frame constraint structure closes (a machine-verified Dirac system with zero propagating frame modes), the nonlocal operator is rigorously defined by a positive Herglotz–Nevanlinna measure, the two-point theory is causal, ghost-free, Cassini-safe and nonlinearly stable, the Einstein-aether strong-coupling wall is mis-applied, and gravitational lensing is viable but tuned**

Carl P. Zimmerman · Briar Creek Tech · 2026-07-08 (v6)

---

### Abstract

The de Sitter–Unruh reading of the MOND acceleration scale,  $a_0 = cH_\Lambda/Z = c^2\sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11} \text{ m s}^{-2}$ , posits that the scale is a modified-**inertia** effect of the cosmic (dark-energy) horizon rather than a modification of gravity. This paper writes the explicit nonlocal modified-inertia action and carries its covariant analysis to a definite boundary, reasoning throughout from the framework’s own premises (a passive frame, with the MOND in the matter sector) rather than the standard aether/MOND lens. Results, each with reproducible verification (both  $a_0$  footings): **(1) the worldline/matter sector is machine-verified** ( $v(y)=\sqrt{(1+1/y)}$  to  $3 \times 10^{-13}$ , Newton, deep-MOND, baryonic Tully–Fisher, the dS-Unruh-forced  $\sqrt{2}$ -DC-weight external-field kernel, exact ghost-freedom). **(2) The modified-inertia realization evades the Cassini quadrupole** by  $\sim 7$  orders (deep-Newtonian  $v \sim 1 \approx 7 \times 10^{-7}$ ), where the modified-gravity realization fails at  $+6$  to  $+14\sigma$ . **(3) Promoting the frame field covariantly, MOND does not force modified gravity** — the frame-equation source is  $l=0$  (soaked by the unit-timelike multiplier) and the matter stress tensor’s *principal* (UV) part is likewise an  $l=0$  isotropic source, so the  $l=2$  Bianchi shear lock that forces AeST’s Cassini quadrupole is not populated at that order; computing the *full* metric stress tensor  $T_{\mu\nu}$  and its subleading multipoles is a named open step. **(4) The mixed matter–aether two-point propagator is causal and ghost-free in closed form** (retarded construction; Källén–Lehmann spectral positivity across the whole cut; principal symbol = the GR light-cone). **(5, correcting v2) The Einstein-aether strong-coupling wall does not wall the framework**: that wall is a property of a *propagating* aether mode whose kinetic norm vanishes, but the framework’s frame is **passive** (its own theory: an algebraic constitutive law, an SME background, the cosmic rest frame), with zero propagating aether modes — so the wall’s object is undefined for it. The wall binds only the rejected AeST/khronon *modified-gravity* limb. **(6) Gravitational lensing — the enhancement is intrinsic to the stress tensor, magnitude open**: varying  $S_{\text{matter}}$  by the metric gives a  $T_{\mu\nu}$  that inherits the nonlocal  $K(\square_u)$ , and in deep-MOND ( $K \approx \sqrt{\square_u/a_0}$ ) an intrinsic,  $a_0$ -enhanced contribution to the very metric photons follow. Two facts are computed: the source is fluid-like, so there is **no gravitational slip** ( $\Psi=\Phi$ , the lensing potential tracks the dynamical one), and the  $a_0$  contribution is the *same order* as the baryonic one ( $\approx 5 \times$  at 20 kpc, not  $v^2/c^2$ -suppressed). Whether it supplies *exactly* the RAR factor  $v(y)$  with the right profile — and does so *without double-counting* the dynamical anomaly — is the one open computation; a bolted-on ghost-condensate remains available as bookkeeping (per-galaxy-tuned) but is a fallback, not a necessity. New in v4, addressing three formal completeness questions: **(7) the passive-frame constraint structure is a machine-verified closed Dirac system**: the full curved-spacetime Dirac analysis terminates at the secondary level (the unit-norm sector is genuinely second-class — its  $2 \times 2$  Dirac block has determinant  $4(u \cdot u)^2 \rightarrow 4$  on-shell — with no tertiary tower), leaving **zero propagating frame degrees of freedom** (2 total: the GR graviton). The crux — that  $u$  sits *inside*  $\square_u$  and might thereby acquire a kinetic term — resolves negatively: the exact quadratic symbol at every order is  $S_n = (-1)^n k^2 \perp k_0 \wedge (2n)$ , whose only root is  $k_0=0$  independent of spatial  $k$ , i.e. a transport-along- $u$  ODE with **no wave-cone and zero group**

**velocity** — the matter coupling does *not* promote  $u$  to a dynamical field. **(8)** The nonlocal operator  $K(\square u)$  is **rigorously defined** by the Borel functional calculus of the (essentially self-adjoint) directional operator  $\square u$ , with a canonical **Herglotz–Nevanlinna spectral representation** carrying a *unique positive Borel measure* (closed-form density, Källén–Lehmann-positive);  $K$  is bounded ( $\|K\| \leq 1$ , operator-monotone) on all of  $L^2$ , causal-retarded, and its non-entire branch cut is the physical de Sitter–Unruh emission continuum, not a ghost. **Nonlinear** classical stability follows from the same no-wave-cone symbol to all orders together with the matter sector *not* modifying the graviton principal symbol — because  $K \rightarrow 1$  in the UV its leading contribution is a non-derivative source, so hyperbolicity is inherited from GR (computed, not assumed); **radiative** stability is structurally protected — the  $(\nabla u)^2$  aether counterterm is symmetry-forbidden by the purely longitudinal  $u$ -derivatives, so  $\alpha_1 = \alpha_2 = 0$  does not run, and  $a_0$  is IR-protected by  $\sqrt{z}$  analyticity plus khronon shift symmetry — but is argued at the power-counting level, not yet by a full covariant one-loop de Sitter computation. **Open:** that covariant one-loop calculation, and the deep-MOND stress-tensor lensing magnitude (whether the nonlocal  $T_{\mu\nu}$  supplies exactly  $v$  without double-counting the dynamics). The MOND sign remains a **postulate** and  $a_0$ 's value is underived. An effective one-scale completion carried to a sharp, named boundary — **not a finished theory and not a theory of everything**.

### 1. Setting and scope

The framework reads  $a_0$  as the acceleration at which a body's inertial response to the de Sitter vacuum departs from Newtonian, with the Deser–Levin temperature  $T(a) = (\hbar/2\pi k_B c) \sqrt{(a^2 + (cH_\Lambda)^2)}$  a horizon floor and the interpolation  $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} a_0)}$ ,  $v(y) = \sqrt{(1+1/y)}$ . The earlier theory-of-everything and Standard-Model claims were publicly retracted (2026-06-23) and are not reasserted; this is a one-scale effective-theory construction with named inputs, each labeled **derived**, **postulated**, or **open**. Crucially — and this is the correction to v2 — the framework is *modified inertia*: the MOND lives in the **matter** sector, and the frame field  $u^\mu$  is a **passive**, horizon-anchored reference (the framework's own paper: an *algebraic constitutive law*  $g_{\text{bar}} = G(a)$  with no frame equation of motion; an SME gravity-sector *background*; the Cassini-constrained cosmic rest frame). It is **not** AeST/Einstein-aether, where a dynamical aether *carries* the MOND — the theory that fails Cassini.

### 2. The action

With signature  $(-+++)$ ,  $S[x, g, u] = S_{\text{EH}}[g] + S_u[g, u] + S_{\text{matter}}[x, g, u]$ : host GR unmodified; a unit-timelike frame constraint  $S_u = -\int \sqrt{-g} (\lambda/2) (u^\mu \mu_{\mu} + 1)$  (no aether kinetic term); and the modified-inertia content in the matter kinetic sector,

$$S_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} \rho_m [s u^\mu K(\square u/a_0^2) u_\mu], \quad K(z) = \frac{\sqrt{1+4z} - 1}{2\sqrt{z}}, \quad \square_u f = u^a \nabla_a (u^b \nabla_b f).$$

$K$  is non-polynomial with a branch cut and a single healthy pole (residue +1). The equivalent Galley doubled-worldline form uses the conservative gradient  $G(a)^{\mu=a} \mu_{\mu} \text{fw}(|a|/a_0)$ ,  $\mu_{\text{fw}}$  the exact inverse of  $v(y) = \sqrt{(1+1/y)}$ ; the external-field kernel  $\theta(y) = \theta_0/(1+(\theta_0-1)y^2)$ ,  $\theta_0 = \sqrt{2}$ , has its shape forced by the de Sitter Wightman function. Here  **$s = -1$  is the MOND-sign postulate**,  $a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11}$  (canonical),  $1.13 \times 10^{-10}$  (alternate footing).

### 3. Verified: the worldline sector and the Cassini evasion

Machine-checked, both footings: the circular-orbit reduction reproduces  $v(y) = \sqrt{(1+1/y)}$  to  $3.35 \times 10^{-13}$ ; Newton, deep-MOND, and baryonic Tully–Fisher  $v^4 = GM/a_0$  (exact); the form-factor  $\leftrightarrow$  RAR identity  $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} a_0)}$ ; the  $\sqrt{2}$ -DC-weight external-field kernel; and ghost-freedom (single pole +1 vs the Ostrogradsky ghost of the local truncation). **Cassini:** the anisotropic galactic external field ( $a_{\text{ext}} = 2.29 a_0$ ) enters only Saturn's inertial response, deep-Newton-suppressed by  $v^{-1} \approx 7 \times 10^{-7}$ ; the  $l=2$  quadrupole is  $\sim 7.4 \times 10^{-34} \text{ s}^{-2} \sim 7$  orders below the  $5.2 \times 10^{-27}$  ceiling. The modified-gravity realization instead fails at +6 to

+14 $\sigma$ . This is Milgrom's [2009] statement that the inner-solar-system quadrupole is an AQUAL/QUMOND effect absent in modified inertia, realized explicitly.

#### 4. The covariant completion, on the framework's own terms

Because the frame is passive and the MOND is in the matter sector, the covariant question is *not* whether a dynamical aether is strongly coupled — it is whether the passive-frame + nonlocal-K matter system is well-posed. Six results:

**(a) MOND does not force modified gravity.** Varying  $S_{\text{matter}}$ , the source it adds to  $u^\mu$ 's field equation is  $J^\mu = -\rho_m s k u^\mu$ :  $l=0$ , parallel to  $u^\mu$ , soaked by the multiplier  $\lambda$ . It never populates the  $l=2$  traceless-shear channel whose divergence is the Bianchi lock that forces AeST's Cassini quadrupole. So the covariant lift does not collapse to modified gravity; a passive frame carries at most  $Q_2^\mu u \lesssim 7 \times 10^{-35} \text{ s}^{-2}$  over an open, ghost-free region. This is established at the *frame-equation* level (the  $\delta S_{\text{matter}}/\delta u$  source) and, for the metric, at *principal-symbol* order: since  $K \rightarrow 1$  in the UV the leading matter stress tensor  $T_{\mu\nu} = -(2/\sqrt{-g})\delta S_{\text{matter}}/\delta g$  is a non-derivative,  $l=0$  isotropic source (`principal_symbol_blockdiag.py`), which cannot populate the  $l=2$  shear. Computing the *full* closed-form nonlocal  $T_{\mu\nu}$  and machine-verifying  $\nabla^\mu T_{\mu\nu} = 0$  on-shell — the direct Bianchi-lock object at subleading order — is the one named open metric-variation step (v4/v5 varied the action w.r.t.  $u$  and  $\lambda$ , not yet the metric); it is a finite classical calculation, not walled.

**(b) The mixed two-point propagator is causal and ghost-free in closed form.** The dynamical- $u$  variation resums to  $Q(z) = K + 2zK' = 2\sqrt{z}/\sqrt{1+4z}$ ; the retarded memory kernel has strictly causal support (an advanced control correctly flips it), Titchmarsh UHP-analyticity holds, and Källén–Lehmann spectral positivity  $\rho(s) \geq 0$  across the whole cut is verified in closed form (a control correctly flags a known Ostrogradsky ghost as  $\rho < 0$ ). The branch cut is a healthy radiation continuum, not a ghost.

**(c) The principal symbol is the GR light-cone,**  $D(w)/w = 1 + i a_0/2w - a_0^2/8w^2 + \dots$ , the nonlocal correction  $O(a_0/w)$  UV-subdominant and IR-gapped — hyperbolic, no secular runaway. So the passive-frame theory is **principal-symbol-well-posed**.

**(d) The Einstein-aether strong-coupling wall is mis-applied (correcting v2).** That wall is, by construction, a property of a *propagating* aether/khronon mode whose canonical kinetic norm vanishes ( $M_{\text{sc}} \sim \sqrt{N} M_{\text{Pl}}$ ; the mode speeds carry the vanishing norm in the denominator) [Blas–Pujolàs–Sibiryakov 2009; Gong et al. 2018]. The framework's frame is passive — zero such modes — so the wall's central object is *undefined*, not violated. It binds only the rejected AeST/khronon modified-gravity limb (which fails Cassini at +6–14 $\sigma$ ). v2's “the covariant route is walled by strong coupling” applied that dynamical-aether wall to the passive-frame realization and is hereby corrected.

**(e) The constraint structure closes as a machine-verified Dirac system (v4).** The full Dirac–Bergmann analysis of the passive-frame + matter system in *curved* spacetime terminates cleanly. The primaries are  $p_\lambda \approx 0$  together with a *degenerate* Legendre map for the  $u$ -momenta (the velocity Hessian is proportional to the spatial  $k^2$ , so there is no invertible hyperbolic kinetic operator — the precise sense in which  $u$  carries no Cauchy data); the secondary is the constraint  $\chi_1: u \cdot u + 1 = 0$ ; and the  $u$ -equation of motion *algebraically* fixes the multiplier  $\lambda = -\rho s$  with **no tertiary tower**. The unit-norm pair  $(\chi_1, u \cdot \pi)$  is genuinely **second-class** — its Dirac block is  $[[0, 2(u \cdot u)], [-2(u \cdot u), 0]]$  with determinant  $4(u \cdot u)^2 = 4$  on-shell (machine-verified carrying the full metric  $g_{\mu\nu}$ ) — so a Dirac bracket exists and the frame-sector degree-of-freedom count is  $\frac{1}{2}[10 - 0 - 10] = 0$ ; the only propagating modes are GR's two graviton polarizations. The crux that  $u$  sits *inside*  $\square_u$  is resolved honestly: the exact khronon variation gives a *nonzero* transverse quadratic form (the naive “identically zero, purely longitudinal” reading is false), but that form is  $S_n = (-1)^n k^2 \perp k_0^{(2n)}$ , whose only nonzero- $k_\perp$  root is  $k_0 = 0$  (a double root independent of spatial  $k$ ) — a transport-along- $u$  ODE with no wave-cone and zero group velocity, **not** a spatially propagating aether kinetic term. Two honest relabelings

surfaced in verification, neither changing the count: hypersurface-orthogonality (twist=0) is the *khronon-frame postulate*, not an algorithm-generated secondary constraint on a generic unit 4-vector; and  $\pi_\mu \approx 0$  is a degenerate Legendre map, not a literal vanishing. The construction is thus mathematically complete **given the khronon-frame premise**.

**(f) The nonlocal operator  $K(\square_u)$  is rigorously defined (v4).** On a globally hyperbolic passive- $u$  background the directional operator  $\square_u = (u \cdot \nabla)^2$  is essentially self-adjoint, so  $K(\square_u) = \int K(\lambda) dE(\lambda)$  is fixed by the Borel functional calculus.  $K(z)$  is a genuine Herglotz–Nevanlinna function of  $z = \square_u / a_0^2$  — proven analytically via the uniformizer  $w = \sqrt{(1+4z)}$  ( $K = \sqrt{(w-1)/(w+1)}$ ) is a composition of upper-half-plane maps) and numerically (min Im  $K$  over the upper half-plane =  $+4.8 \times 10^{-6}$ , zero violations). The Herglotz–Nevanlinna representation theorem then supplies a *unique positive Borel measure* with closed-form density ( $\rho_A = (1 - \sqrt{(1-4|t|)}) / (2\pi\sqrt{|t|})$  on  $-1/4 < t < 0$ ,  $\rho_B = 1 / (2\pi\sqrt{|t|})$  on  $t < -1/4$ ), everywhere  $\geq 0$  (Källén–Lehmann), with  $\int d\mu / (1+t^2)$  finite; the representation reconstructs  $K$  to  $< 10^{-11}$ .  $K(\square_u)$  is bounded ( $\|K\| \leq 1$ , operator-monotone/Loewner), defined on all of  $L^2$ , and causal-retarded. The non-entire branch cut is the physical de Sitter–Unruh emission-threshold continuum, not an obstruction. One relabeling improves the physical reading:  $K(\square_u)$  is self-adjoint on the static  $z > 0$  sector, while on the dynamical  $z < 0$  (oscillatory) sector the spectrum lies on the cut and  $K(\square_u)$  is **normal** and causal-retarded (its positive anti-Hermitian part is the Källén–Lehmann-guaranteed dissipative response), not globally self-adjoint.

**What is open.** Nonlinear classical stability now follows from (e): the exact symbol  $S_n = (-1)^n k^2 \perp k_0^{2n}$  holds at every order, background gradients and curvature add strictly lower-derivative pieces that cannot flip the cone, so there is no nonlinear ghost or runaway and hyperbolicity is inherited from GR — this last step now *computed*, not assumed: because  $K \rightarrow 1$  in the UV the matter sector’s principal part is non-derivative, so its stress tensor cannot enter the graviton kinetic symbol (principal\_symbol\_blockdiag.py;  $K \rightarrow 0$  in the IR, both bounded). What remains open is **radiative** stability as a *computation*: the  $(\nabla u)^2$  aether-kinetic counterterm is symmetry-forbidden (every  $u$ -derivative in every vertex is longitudinal,  $u \cdot \nabla$ , so no pure-transverse piece can be generated —  $\alpha_1 = \alpha_2 = 0$  is not a tuning that runs),  $a_0$  is IR-protected by the  $\sqrt{z}$  IR-analyticity (no  $z^0$  tadpole) plus khronon shift symmetry  $T \rightarrow T + \text{const}$ , and Källén–Lehmann positivity survives loops by the pump-free/KMS argument — but a full covariant background-field one-loop calculation on de Sitter (confirming no  $(\nabla u)^2$  counterterm, the  $a_0$ -vertex decoupling, and dressed positivity from a real loop integral) is not performed. Two finite, unwallled *classical* computations are the natural next step: the full closed-form nonlocal stress tensor  $T_{\mu\nu} = -(2/\sqrt{-g}) \delta S_{\text{matter}} / \delta g$  with machine-verified on-shell conservation, and the coupled nonlinear Cauchy problem for the branch-cut class built on it. A first-principles ghost-condensate profile law (§5) also remains open.

## 5. Gravitational lensing: the enhancement is intrinsic to the stress tensor (magnitude open)

In modified inertia the modification is to how massive bodies respond, so it is tempting to conclude that light — massless, with no inertia to modify — sees only the baryonic metric, and that lensing must be supplied by a separate dark sector placed by hand. That conclusion is premature. Varying the covariant matter action by the metric,  $T_{\mu\nu} = -(2/\sqrt{-g}) \delta S_{\text{matter}} / \delta g$  inherits the nonlocal operator: the connection variation  $\delta \Gamma$ , integrated by parts, dresses the baryonic source  $\rho_{\mu\nu}$  with  $K'(\square_u)$ , and in the deep-MOND branch  $K \approx \sqrt{(\square_u)/a_0}$  this dressing is  $O(a_0)$  and *nonlocal* — an intrinsic,  $a_0$ -enhanced contribution to the very metric photons follow (mi\_lensing\_from\_stress\_tensor.py). Two structural facts are computed. First, the source is fluid-like (built from  $u_\mu u_\nu$  and  $g_{\mu\nu}$ ): **no anisotropic stress  $\rightarrow$  no gravitational slip  $\rightarrow \Psi = \Phi$** , so the lensing potential *tracks* the dynamical one — AeST’s no-slip property, here recovered from modified inertia with no Cassini-failing aether. Second, the  $a_0$  contribution to the lensing potential,  $a_0 r / c^2$ , is the *same order* as the baryonic one ( $\approx 5 \times$  at 20 kpc), not  $v^2/c^2$ -suppressed — the enhancement fuel is intrinsic and correctly sized. **Open — the decisive calculation:** whether the closed-form nonlocal  $T_{\mu\nu}$  in

deep-MOND supplies *exactly* the RAR factor,  $(\Phi+\Psi)/2g_{\text{bar}}=v(y)$ , with the right radial profile, *and* does so without double-counting the dynamical anomaly (an  $a_0$ -enhanced source that also drove the rotation curve would over-produce it; consistency requires light to see  $v$  while the inertia-modified dynamics are driven by the baryonic  $g$ ). This is a finite classical computation, its target set by the observed weak-lensing RAR = dynamical RAR [Brouwer et al. 2021; Mistele & McGaugh 2024]. A bolted-on ghost-condensate sector ( $M_{\text{cond}}=M_{\text{bar}}(v(y)-1)$ ) remains available as *bookkeeping* for the same enhancement — viable but per-galaxy-tuned, a free shift-charge integration constant ( $dI_0/da_0=0$ ) — but it is a fallback, not a necessity: the framework’s own stress tensor is the primary, unforced route. Verdict: no slip and an intrinsic, correctly-sized enhancement are computed; the magnitude, profile, and no-double-counting are open — lensing is neither solved nor shown to require dark matter. The genuinely distinctive modified-inertia signature lives in the *dynamical* sector (the MG-impossible non-adiabatic velocity-dispersion hysteresis), not in lensing.

## 6. Honest standing

Reasoning from the framework’s own passive-frame, MOND-in-matter premise: the worldline sector is written and verified; the modified-inertia realization evades Cassini; MOND does not force modified gravity (at the frame-equation and principal-symbol level); the mixed two-point propagator is causal, ghost-free (Källén–Lehmann), and principal-symbol-well-posed; the Einstein-aether strong-coupling wall is a mis-application that does not bind the framework (it binds only the rejected modified-gravity limb); and — new in v4 — the passive-frame **constraint structure closes as a machine-verified Dirac system with zero propagating frame modes**, the nonlocal operator  $K(\square_u)$  is **rigorously defined** by a unique positive Herglotz–Nevanlinna measure, and **nonlinear classical stability holds** (the no-wave-cone symbol to all orders). Gravitational lensing has no gravitational slip and an intrinsic, correctly-sized  $a_0$  enhancement inside its own stress tensor (not a bolted-on sector); whether that  $T_{\mu\nu}$  supplies exactly  $v$  without double-counting the dynamics is open — not solved, not a kill, and not shown to require dark matter. The completion is therefore **not** the modified-gravity limb (Cassini-failing) and **not yet** a finished field theory — it is a well-posed worldline theory whose passive-frame covariantization has a closed constraint algebra (0 frame dof), a rigorously defined causal-retarded nonlocal operator, a causal ghost-free two-point sector, principal-symbol well-posedness, and nonlinear stability, with three honest open edges: the full closed-form metric stress tensor  $T_{\mu\nu}=-(2/\sqrt{-g})\delta S_{\text{matter}}/\delta g$  with on-shell conservation (and the coupled nonlinear Cauchy problem built on it), a full covariant one-loop de Sitter computation certifying the (structurally protected) radiative stability, and the deep-MOND stress-tensor lensing computation (does the nonlocal  $T_{\mu\nu}$  supply exactly  $v$ , with the right profile and no double-counting of the dynamics). The MOND sign stays postulated and  $a_0$ ’s value underived. No completeness or theory-of-everything claim is made.

## 7. Related work

The construction sits between the relativistic modified-gravity realizations of MOND — AeST [Skordis & Złośnik 2021], the Blanchet–Skordis khronon [2025], the nonlocal mimetic model of Deffayet & Woodard [2026] — and the modified-inertia program of Milgrom [1999, 2009]. The covariant frame sector is Einstein-aether [Jacobson & Mattingly 2001], its PPN structure [Foster & Jacobson 2006], its GW170817 constraint [Gong et al. 2018], and its strong-coupling corner [Blas, Pujolàs & Sibiryakov 2009]; the nonlocal well-posedness class is Källén–Lehmann/Herglotz–Nevanlinna (passive-causal), distinct from the entire-function class [Biswas et al.; Barnaby & Kamran]. The doubled-worldline formalism is Galley [2013]; the lensing constraint is Brouwer et al. [2021] and Mistele & McGaugh [2024].

## 8. Reproducibility

Committed scripts (repository `real_research/reviews/`): the worldline sector and Cassini evasion (`cassini_mi_evasion_2026/`); the covariant kinetic wins (`mi_kinetic_completion_2026/`); the mixed-propagator causality + Källén–Lehmann positivity (`mi_propagator_2026/`); the

framework-first correction of the strong-coupling wall (mi\_strongcoupling\_framework\_2026/); the all-orders Cauchy checks (mi\_cauchy\_wellposed\_2026/); the lensing fine-tuning analysis (mi\_lensing\_2026/); and — new in v4 — the closed Dirac constraint analysis, the no-wave-cone crux symbol, the Herglotz–Nevanlinna operator definition, and the nonlinear/radiative stability analysis, and the graviton principal-symbol block-diagonality via the  $K \rightarrow 1$  UV limit (mi\_formal\_completion\_2026/: A10\_dirac\_block.py, A6b\_higher\_n.py, constraint\_structure.py, crux\_variation\_check.py, operator\_definition.py, stability.py, transverse\_mode\_analysis.py, principal\_symbol\_blockdiag.py, mi\_lensing\_from\_stress\_tensor.py). Sign-wall inputs: ACTIVE\_KERNEL\_SIGNTHEOREM\_2026-06.md, FOURTH\_ORDER\_SIGN\_WALL\_2026-07.md; trichotomy: COVARIANT\_MI\_COMPLETION\_2026-06.md.

## References

1. M. Milgrom, Phys. Lett. A **253**, 273 (1999); MNRAS **399**, 474 (2009).
2. S. Deser & O. Levin, Class. Quantum Grav. **14**, L163 (1997).
3. T. Jacobson & D. Mattingly, Phys. Rev. D **64**, 024028 (2001); B. Z. Foster & T. Jacobson, Phys. Rev. D **73**, 064015 (2006), gr-qc/0509083.
4. D. Blas, O. Pujolàs & S. Sibiryakov, JHEP **10**, 029 (2009), arXiv:0906.3046.
5. C. Skordis & T. Złotnik, Phys. Rev. Lett. **127**, 161302 (2021); L. Blanchet & C. Skordis, arXiv:2507.00912 (2025).
6. C. Deffayet & R. P. Woodard, JCAP **04**, 081 (2026), arXiv:2512.10513.
7. Y. Gong, S. Hou, D. Liang & E. Papantonopoulos, Phys. Rev. D **97**, 124023 (2018), arXiv:1802.04303.
8. R. S. Park et al., arXiv:2602.17884 (2026); DESI Collaboration, arXiv:2503.14738 (2025).
9. N. Barnaby & N. Kamran, JHEP **02**, 008 (2008); T. Biswas et al., JHEP **10**, 054 (2010).
10. M. Brouwer et al., A&A **650**, A113 (2021), arXiv:2106.11677; T. Mistele & S. McGaugh, arXiv:2310.15248 (2024).
11. C. R. Galley, Phys. Rev. Lett. **110**, 174301 (2013).
12. P. A. M. Dirac, *Lectures on Quantum Mechanics* (Yeshiva Univ., 1964); M. Henneaux & C. Teitelboim, *Quantization of Gauge Systems* (Princeton, 1992).
13. Herglotz–Nevanlinna representation and the Borel functional calculus: M. Reed & B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis* (Academic, 1980); F. Gesztesy & E. Tsekanovskii, Math. Nachr. **218**, 61 (2000).
14. Companion (this program): sign wall, Zenodo 10.5281/zenodo.21184373; covariant-MI quadrichotomy, 10.5281/zenodo.21148494; written MI action v1/v2/v3/v4/v5, 10.5281/zenodo.21253645 / 21260830 / 21263846 / 21264727 / 21265804.

\*Both  $a_0$  footings throughout. Each claim labeled derived / postulated / open. New in v4–v5: the passive-frame constraint structure closes as a machine-verified Dirac system (0 propagating frame dof), the nonlocal operator  $K(\square_u)$  is rigorously defined by a unique positive Herglotz–Nevanlinna measure, and nonlinear classical stability holds — the graviton-cone step now computed via the  $K \rightarrow 1$  UV limit. The “MOND does not force modified gravity” claim is established at the frame-equation and principal-symbol level; the full metric stress tensor  $T_{\mu\nu}$  and the coupled nonlinear Cauchy problem are named open classical computations alongside the covariant one-loop de Sitter edge. New in v6: gravitational lensing has no gravitational slip and an intrinsic, correctly-sized  $a_0$  enhancement inside the framework’s own stress tensor (not a bolted-on sector) — whether it supplies exactly  $v$  without double-counting the dynamics is the open deep-MOND  $T_{\mu\nu}$  computation; the ghost-condensate is bookkeeping, not a necessity. The MOND sign is a postulate; the Einstein-aether wall is a mis-application. No completeness or theory-of-everything claim.\*